

SHETH LUJ AND SIR MV COLLEGE
Subject: Data Analysis with SAS / SPSS / R

M2 Practical No: 1

Aim: Generating descriptive statistics using `summary()` or `describe()` (R).

Code:

1. Install Required Packages (Run only once)

2. Load Libraries

```
library(titanic)
library(dplyr)
```

3. Import Dataset

```
titanic_data <- titanic_train
```

4. View Dataset Structure

```
print("--- 4. Data Structure (First 6 Rows) ---")
print(head(titanic_data))
```

5. Frequency Table using `table()` (Base R)

```
print("--- 5. Frequency Table: Passenger Class (Pclass) ---")
class_table <- table(titanic_data$Pclass)
print(class_table)
```

```
print("--- 5. Relative Frequency Table: Passenger Class ---")
class_prop_table <- prop.table(class_table)
print(class_prop_table)
```

6. Frequency Table using `count()` (Tidyverse/dplyr)

```
print("--- 6. Frequency Table: Sex (Gender) using dplyr::count ---")
sex_count <- titanic_data %>%
  count(Sex, sort = TRUE) # sort = TRUE orders by frequency (n)
print(sex_count)
```

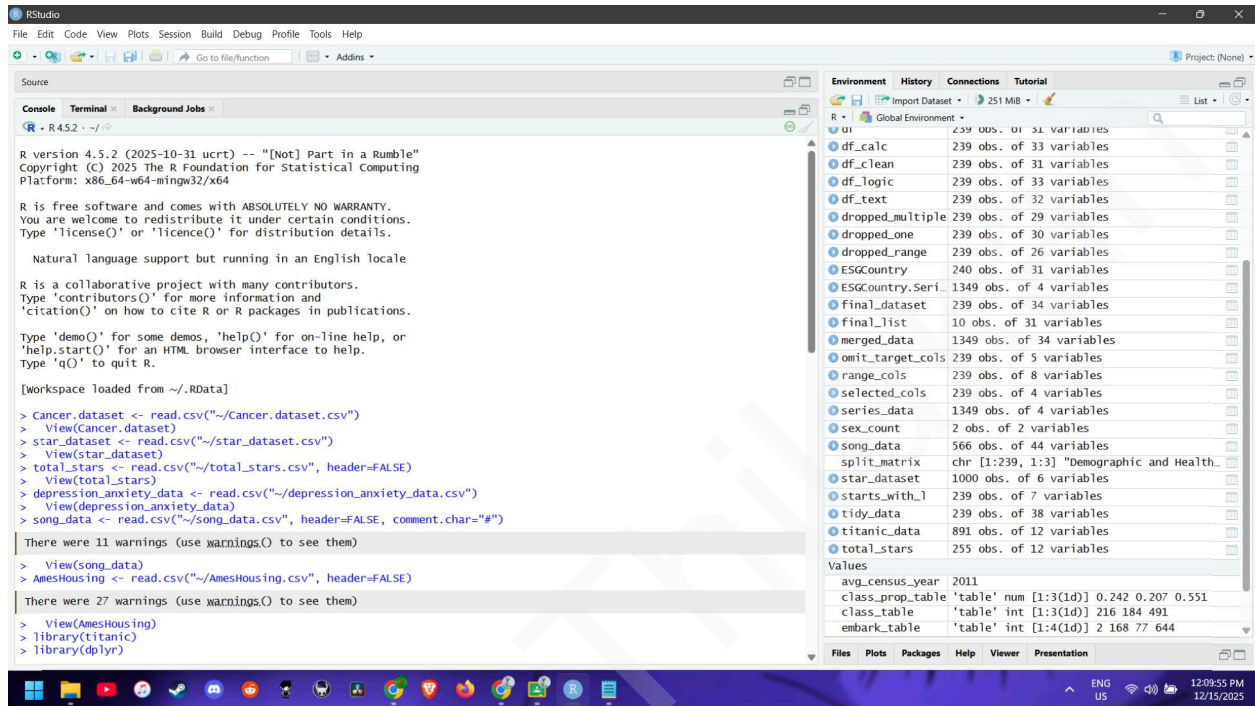
7. Frequency Table for another variable (Embarked)

```
print("--- 7. Frequency Table: Port of Embarkation (Embarked) ---")
```

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```
embark_table <- table(titanic_data$Embarked, useNA = "ifany") # useNA = "ifany" shows NA count  
print(embark_table)
```

Output:



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